

Task 1: ML Classification Project

Alfido Tech | Artificial Intelligence Internship

Internship domain: Artificial Intelligence

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1. Links

- LinkedIn profile: <https://www.linkedin.com/in/ezaz-ahamad-440416336>
- GitHub repository: <https://github.com/ahmed-web-oss-wj/alfido-task1-ml-classification>
- Notebook: ml_classification.ipynb (in the repository root)
- README with environment setup and run commands: README.md (in the repository root)

2. Problem and dataset

The goal is to build and evaluate a supervised classification model that predicts whether a breast tumour is malignant or benign from measurements of cell nuclei. The dataset is the Breast Cancer Wisconsin (Diagnostic) dataset bundled with scikit-learn: 569 samples, 30 numeric features, no missing values and no duplicate rows. The classes are moderately imbalanced (62.7% benign, 37.3% malignant).

3. Approach

- Preprocessing: 80/20 stratified train/test split (455 train, 114 test) so both sets keep the class ratio.
- Scaling: StandardScaler placed inside a scikit-learn Pipeline, so it is fitted only on training folds and cannot leak test information.
- Cross-validation: 5-fold stratified CV on the training set for every model.
- Algorithms compared: Logistic Regression, Random Forest and Gradient Boosting.
- Tuning: GridSearchCV optimising ROC-AUC for Logistic Regression (C) and Random Forest (trees, depth, leaf size).
- Metrics: accuracy, precision, recall, F1 and ROC-AUC, all reported on the untouched test set.

4. Exploratory analysis

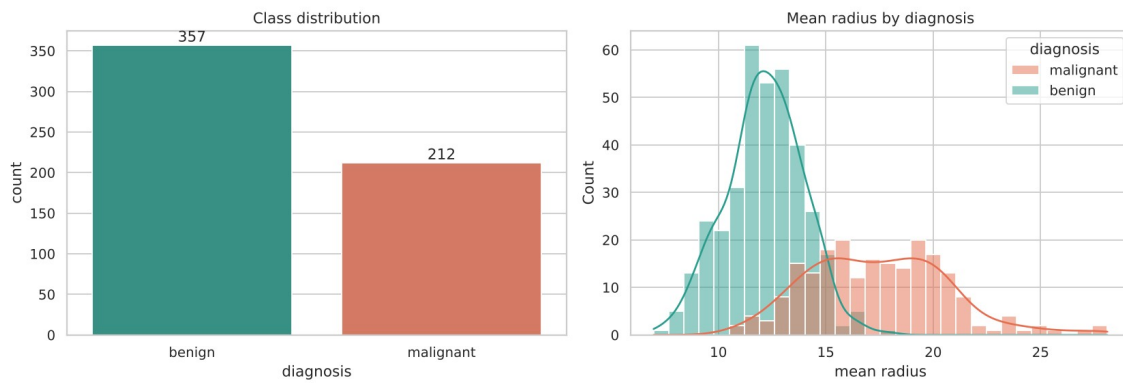


Figure 1: Class distribution and mean radius by diagnosis

Size and shape features such as radius, perimeter, area and concavity separate the two classes well, which is why even a linear model performs strongly.

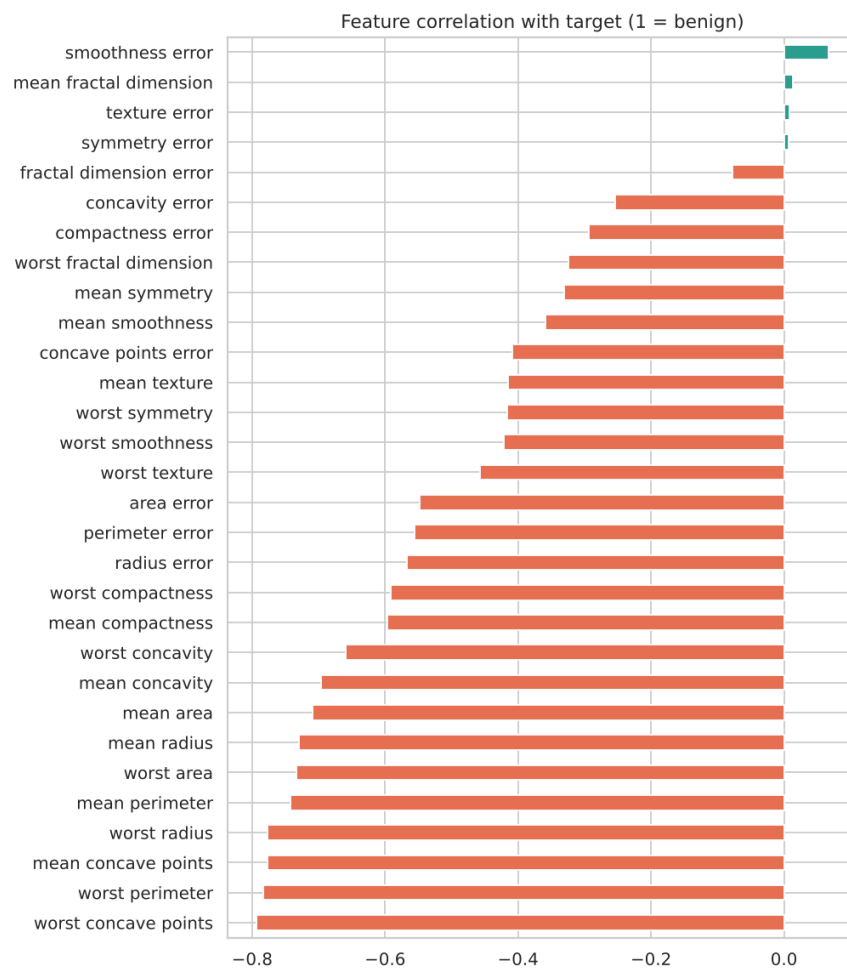


Figure 2: Correlation of each feature with the target

5. Results

Test-set performance of all models (114 held-out samples):

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9825	0.9861	0.9861	0.9861	0.9954
Random Forest	0.9474	0.9583	0.9583	0.9583	0.9937
Gradient Boosting	0.9561	0.9467	0.9861	0.9660	0.9907
Logistic Regression (tuned)	0.9825	0.9861	0.9861	0.9861	0.9954
Random Forest (tuned)	0.9561	0.9589	0.9722	0.9655	0.9934

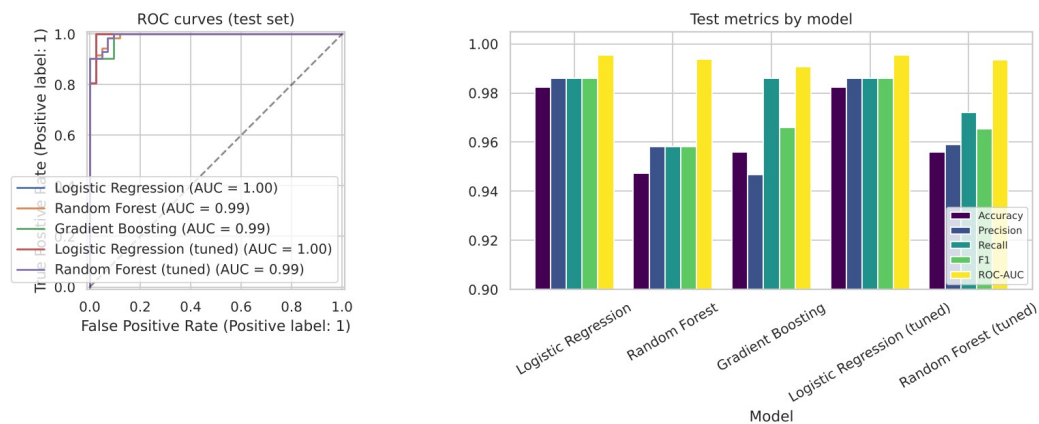


Figure 3: ROC curves and metric comparison on the test set

6. Model selection

Logistic Regression was selected. It scored the highest F1 (0.9861) and ROC-AUC (0.9954) on the test set and also had the best cross-validated ROC-AUC (about 0.996), so the result is not a lucky split. Tuning changed nothing for it (the default $C=1$ was already best), while tuning improved the Random Forest slightly (accuracy 0.9474 to 0.9561) but not enough to overtake the linear model. A simple, fast, interpretable model winning here is a common outcome on small, clean, well-separated tabular data.

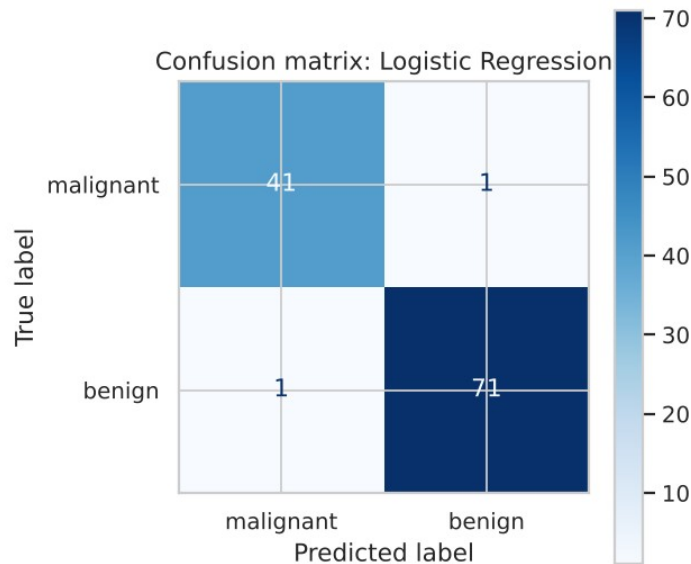


Figure 4: Confusion matrix of the selected model

Of 114 test samples the model made 2 errors: 1 malignant tumour predicted as benign (false negative) and 1 benign tumour predicted as malignant (false positive). Recall for the malignant class is 0.976.

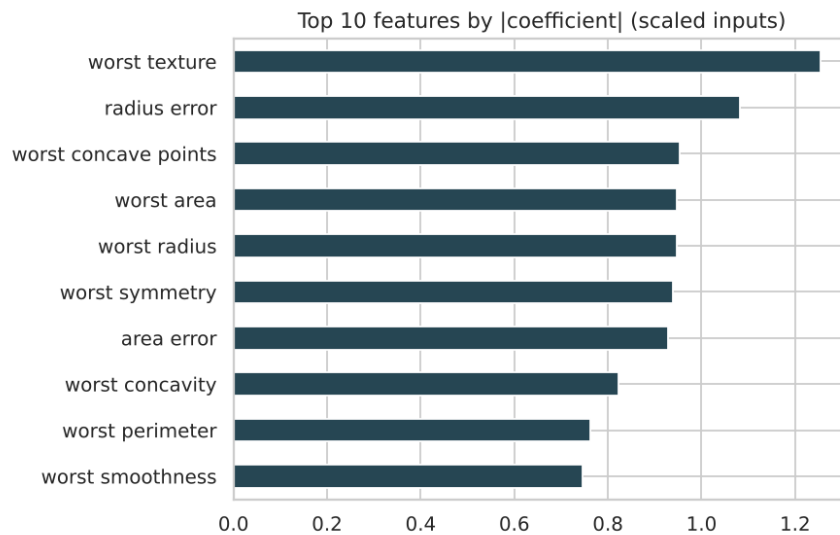


Figure 5: Most influential features

7. Conclusions and limitations

- All models reach roughly 95-98% accuracy and above 0.99 ROC-AUC; the differences between them are small.

- In medical screening a missed cancer (false negative) costs more than a false alarm, so recall on the malignant class is the metric to watch. A decision threshold below 0.5 could trade some precision for higher recall.
- The dataset is small (569 rows) and one 114-sample test set is noisy: a single extra error changes accuracy by about 0.9 points. Results would need external validation before any real use.
- Next steps: probability calibration, threshold tuning, and model interpretation with SHAP (covered in Task 4).